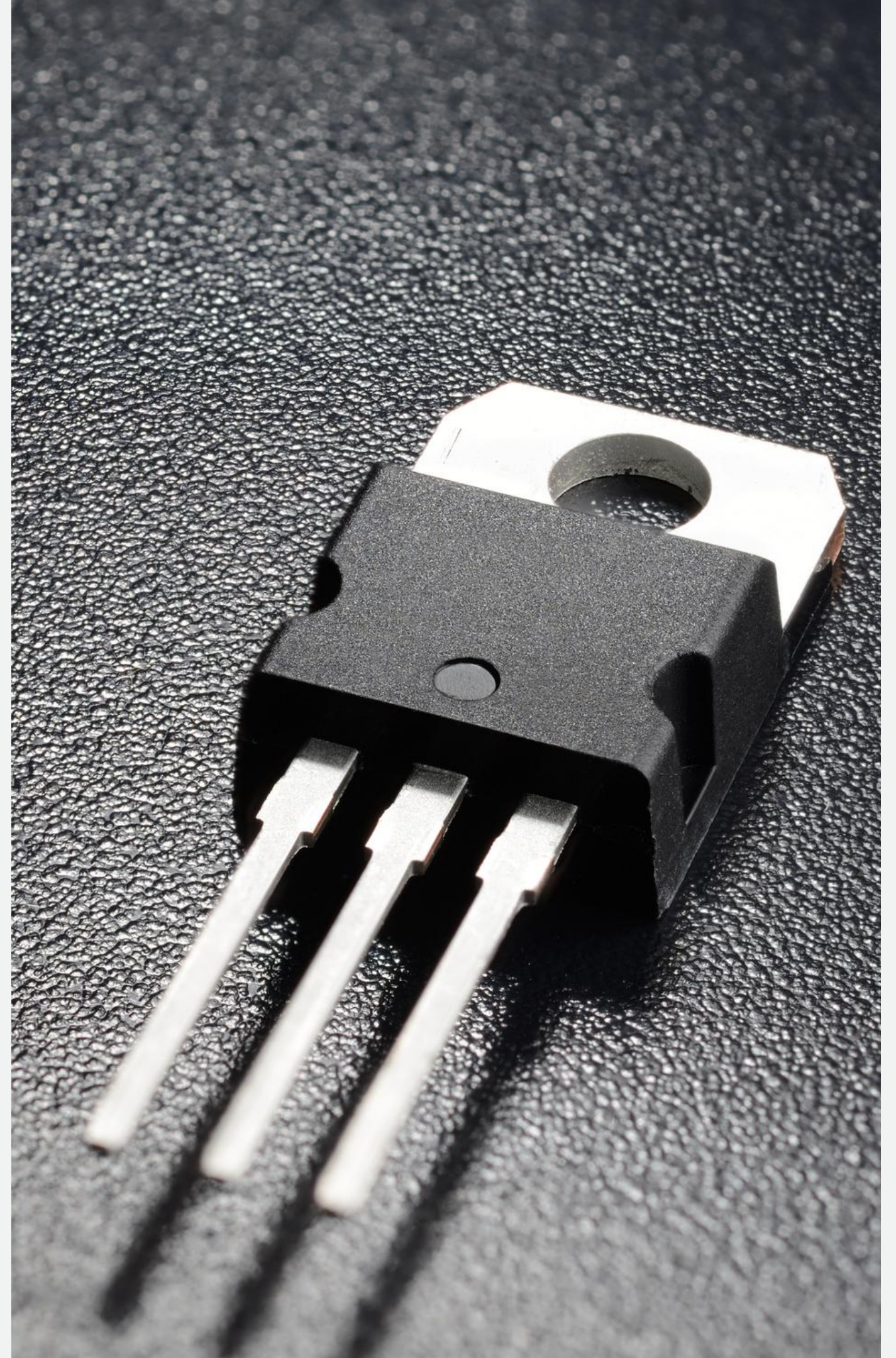


SOLID STATE MATERIALS

TUNNELING FIELD EFFECT TRANSISTORS

GROUP A



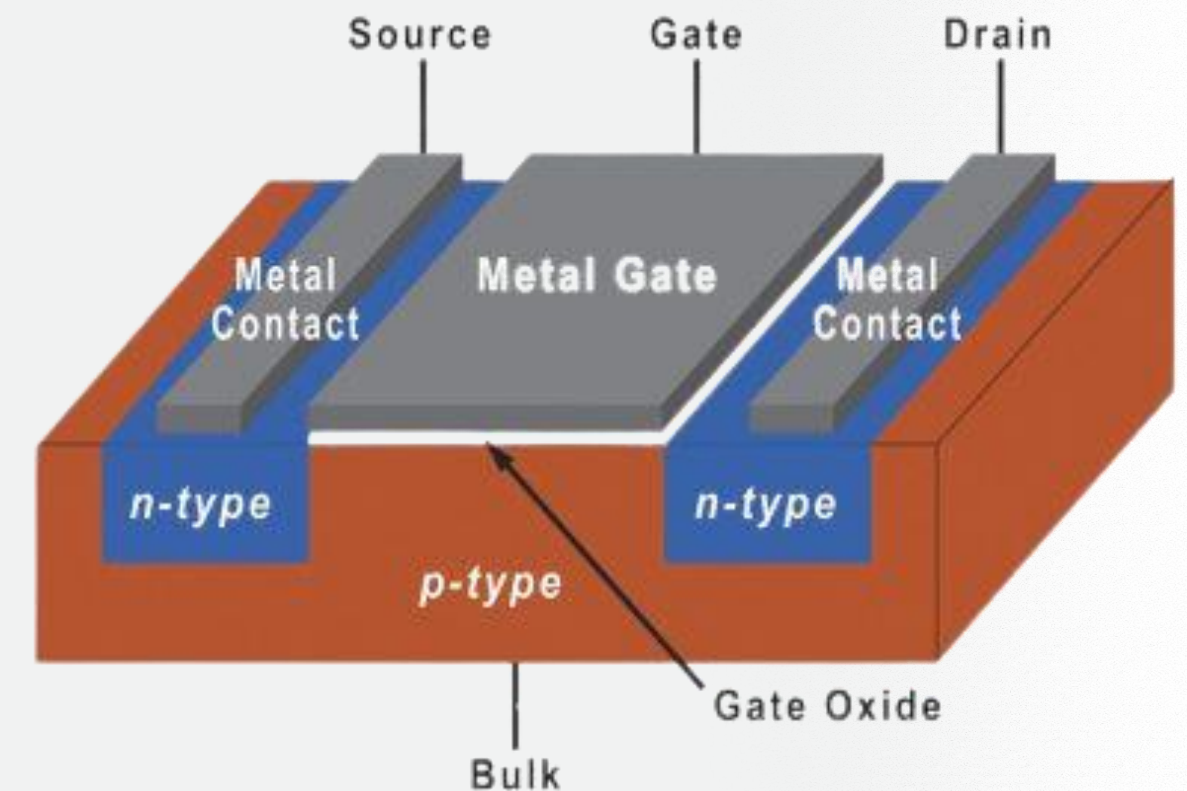
How Transistors Power the Modern World

- FUNDAMENTAL BUILDING BLOCK
- MINIATURIZATION
- SWITCHING & AMPLIFICATION

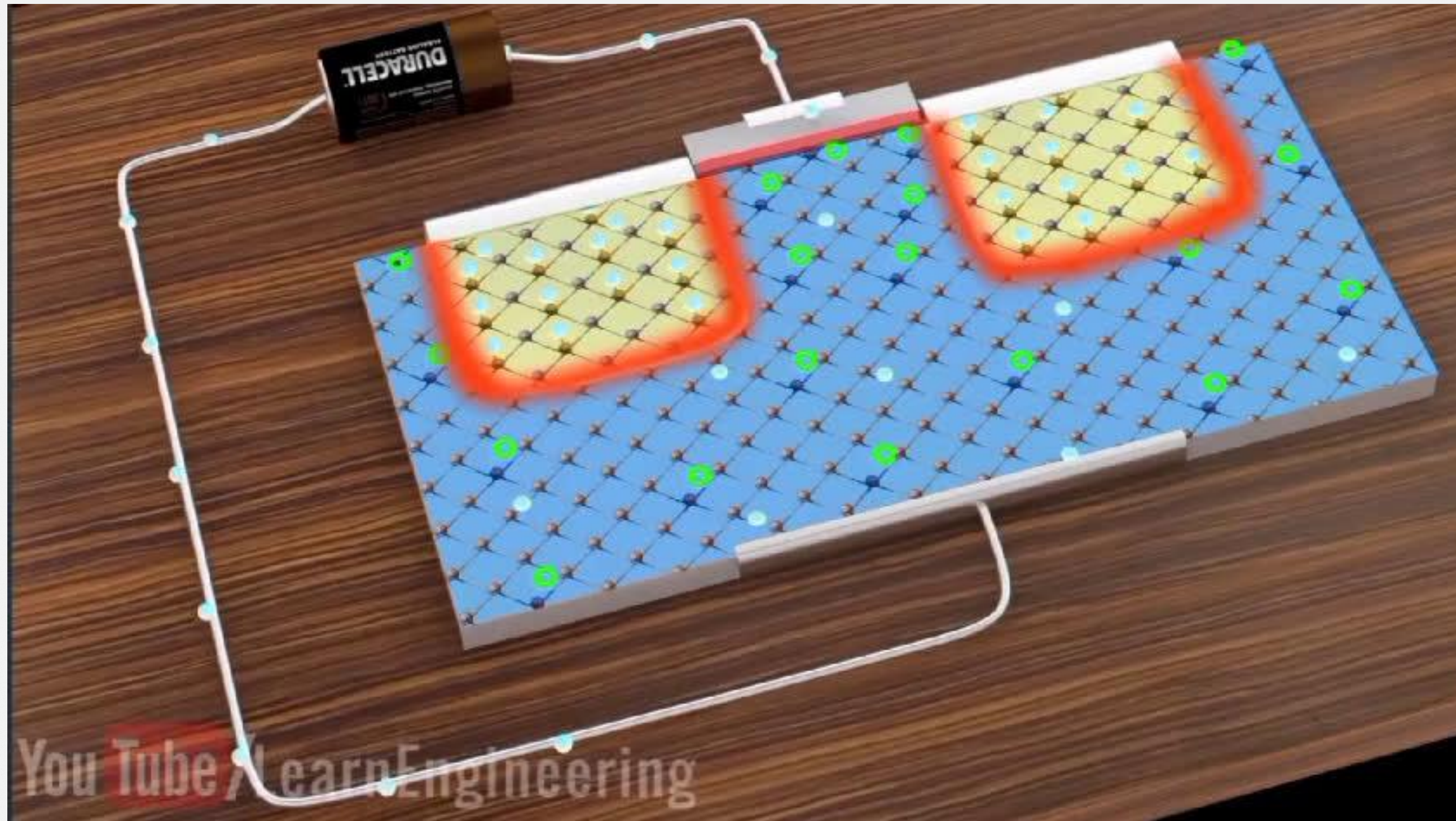


MOSFET

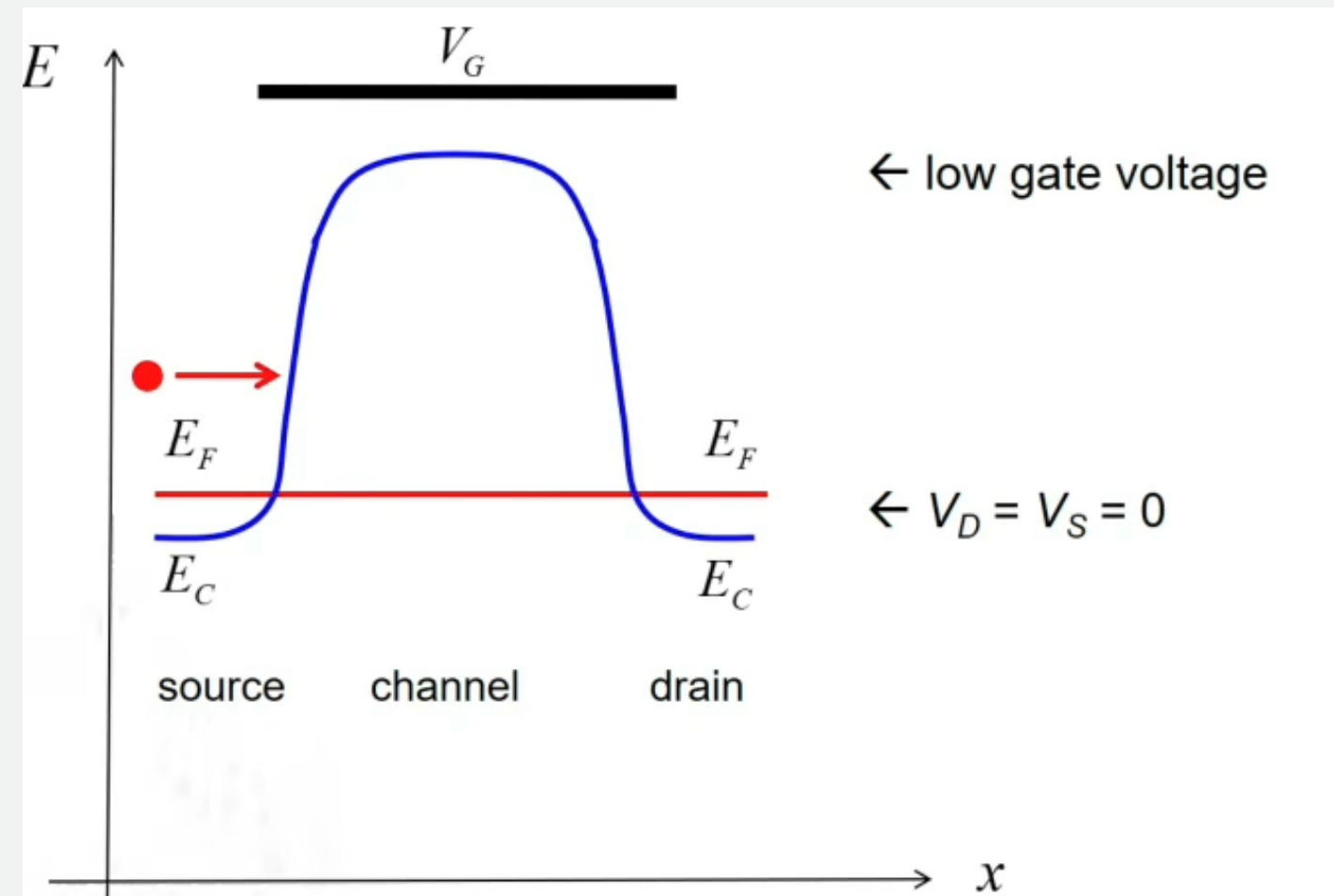
- MOST COMMON AND VERSATILE TRANSISTOR
- EFFICIENT SWITCHING PROCESS
- HIGHER THERMAL STABILITY
- NO INPUT CURRENT REQUIRED



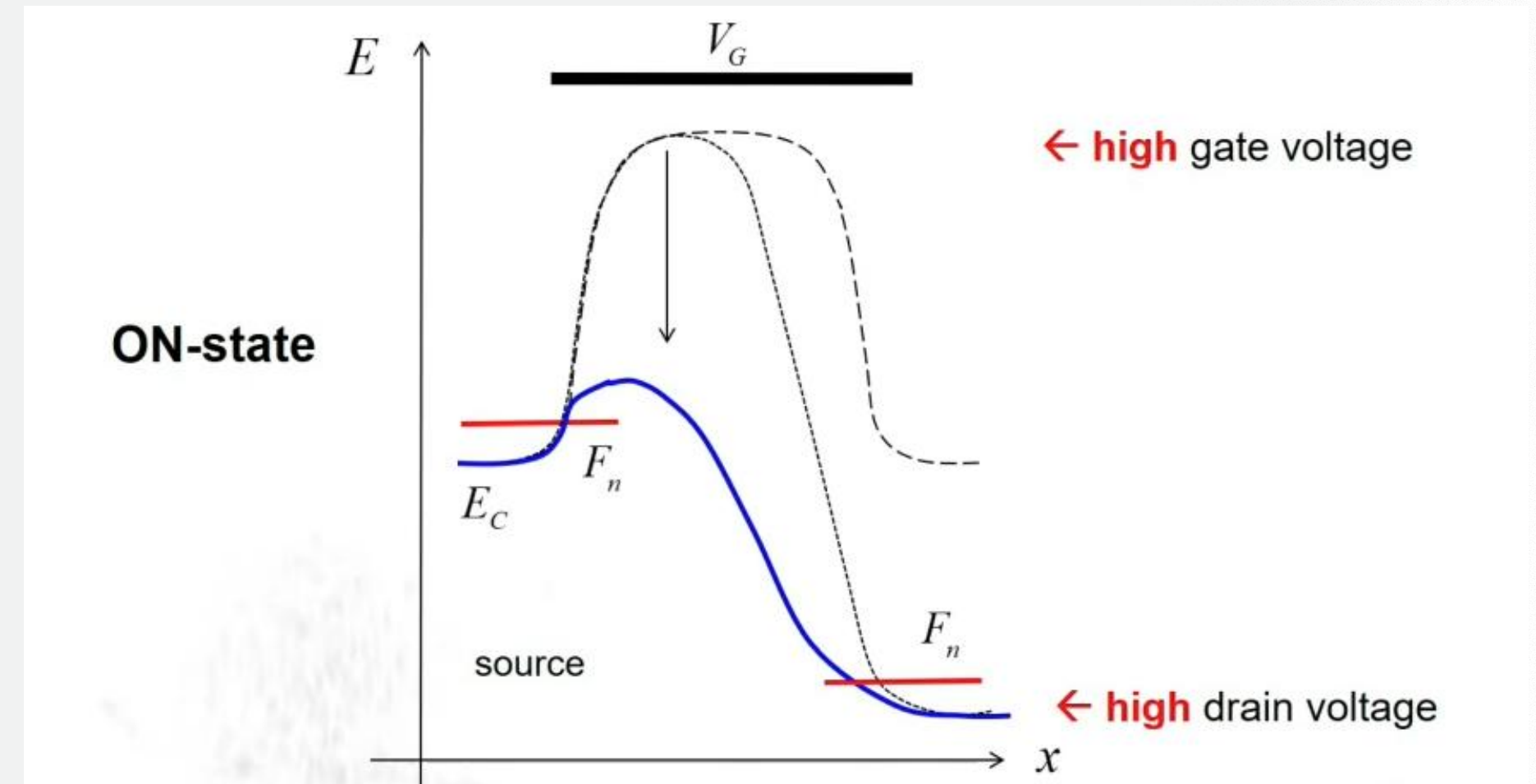
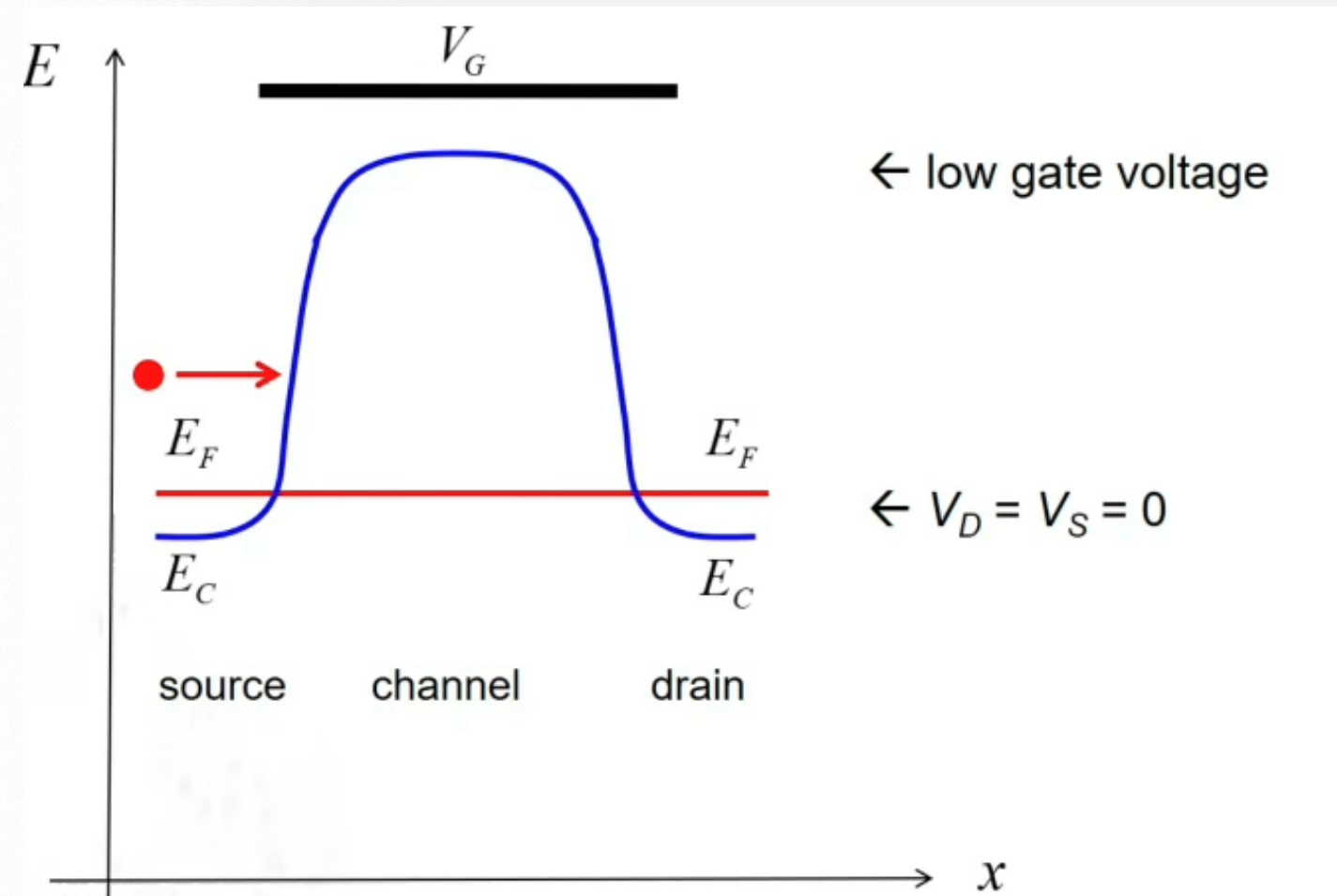
ACTIVATION PROCESS OF MOSFET



BANDGAP DIAGRAM OF MOSFET

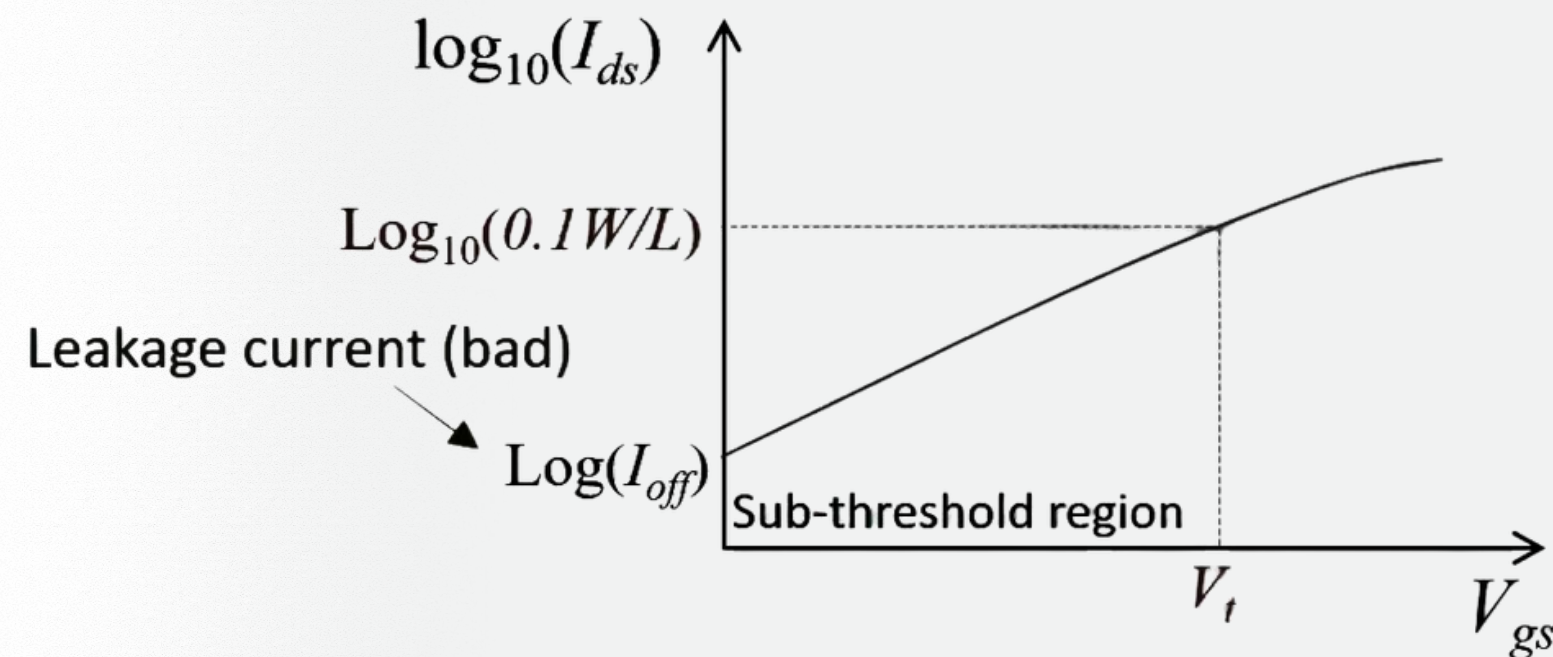


BANDGAP DIAGRAM OF MOSFET



LIMITATIONS OF MOSFET

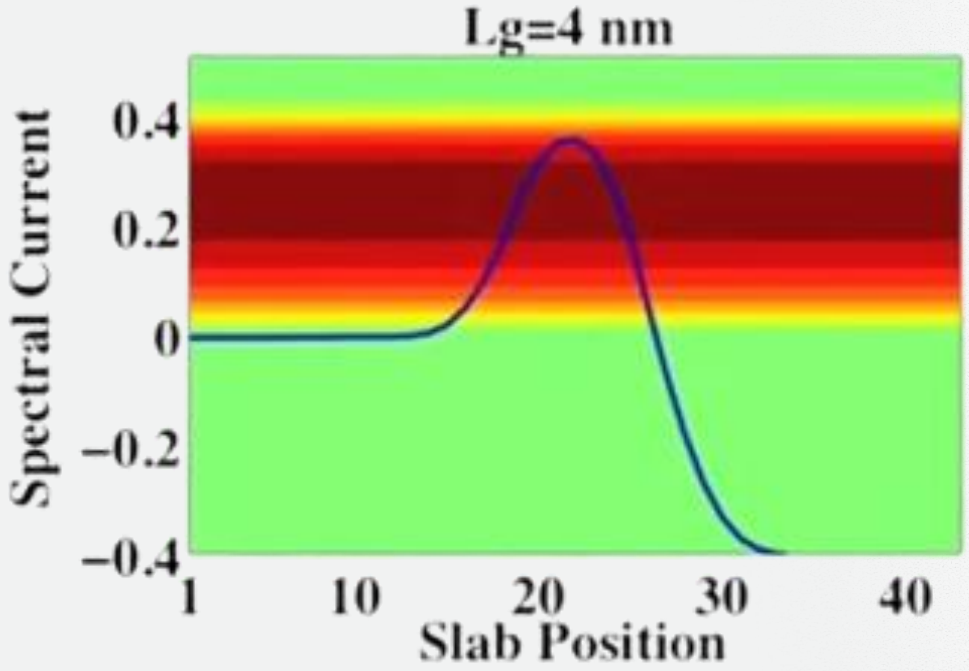
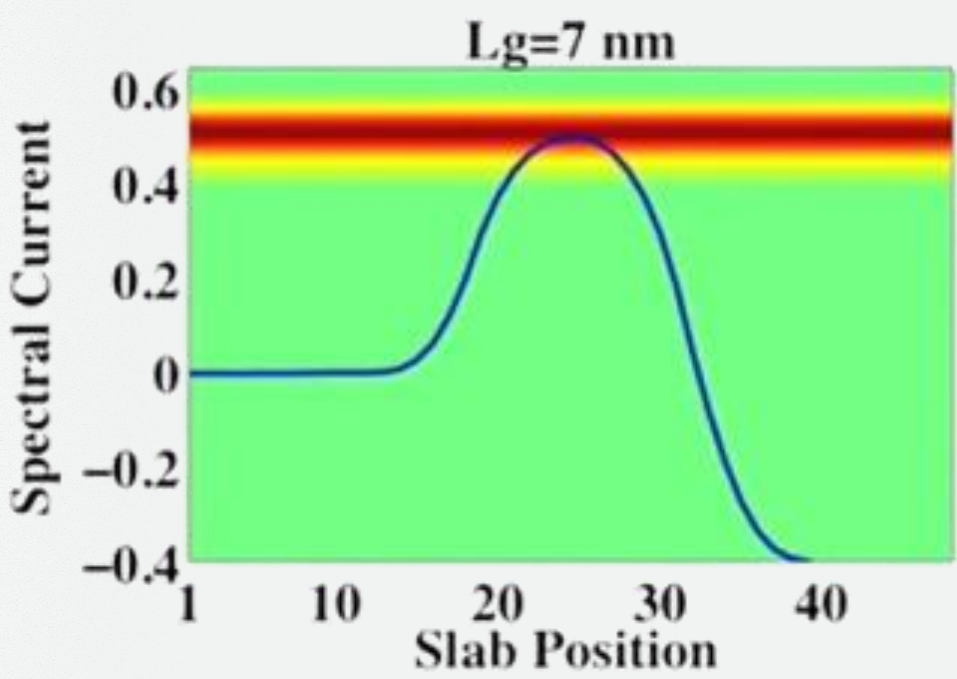
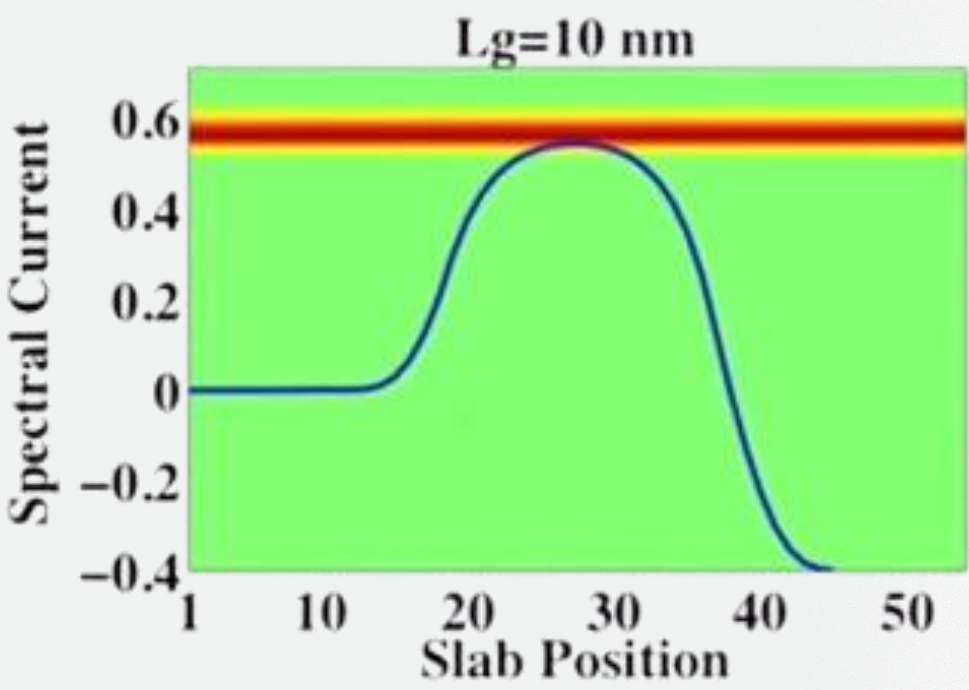
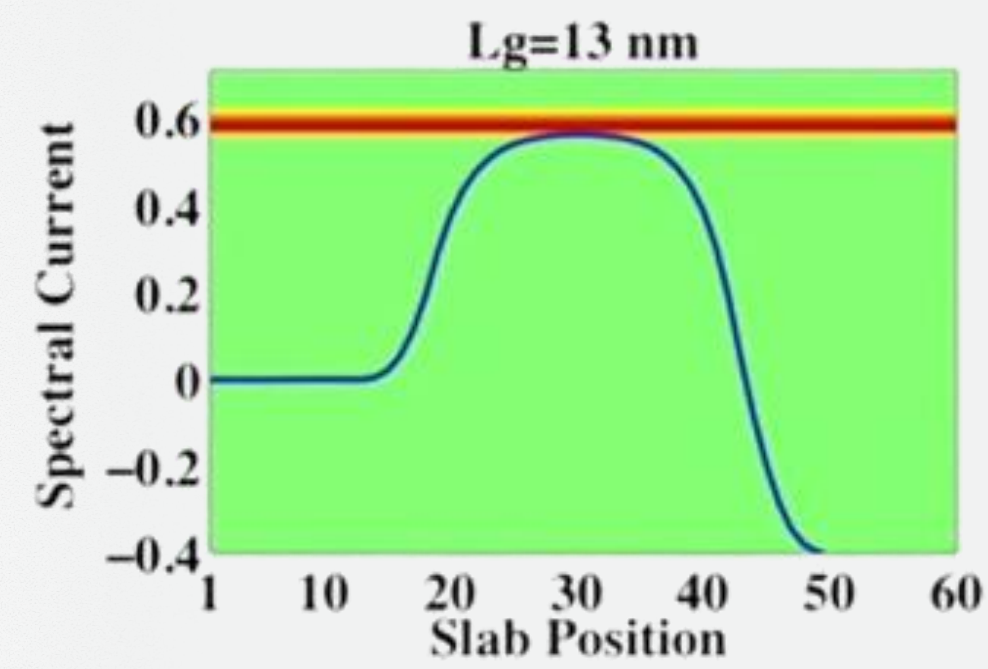
- THE SUBTHRESHOLD SWING OF MOSFETS' CAN'T BE REDUCED BELOW 60mV/DEC AT ROOM TEMPERATURE



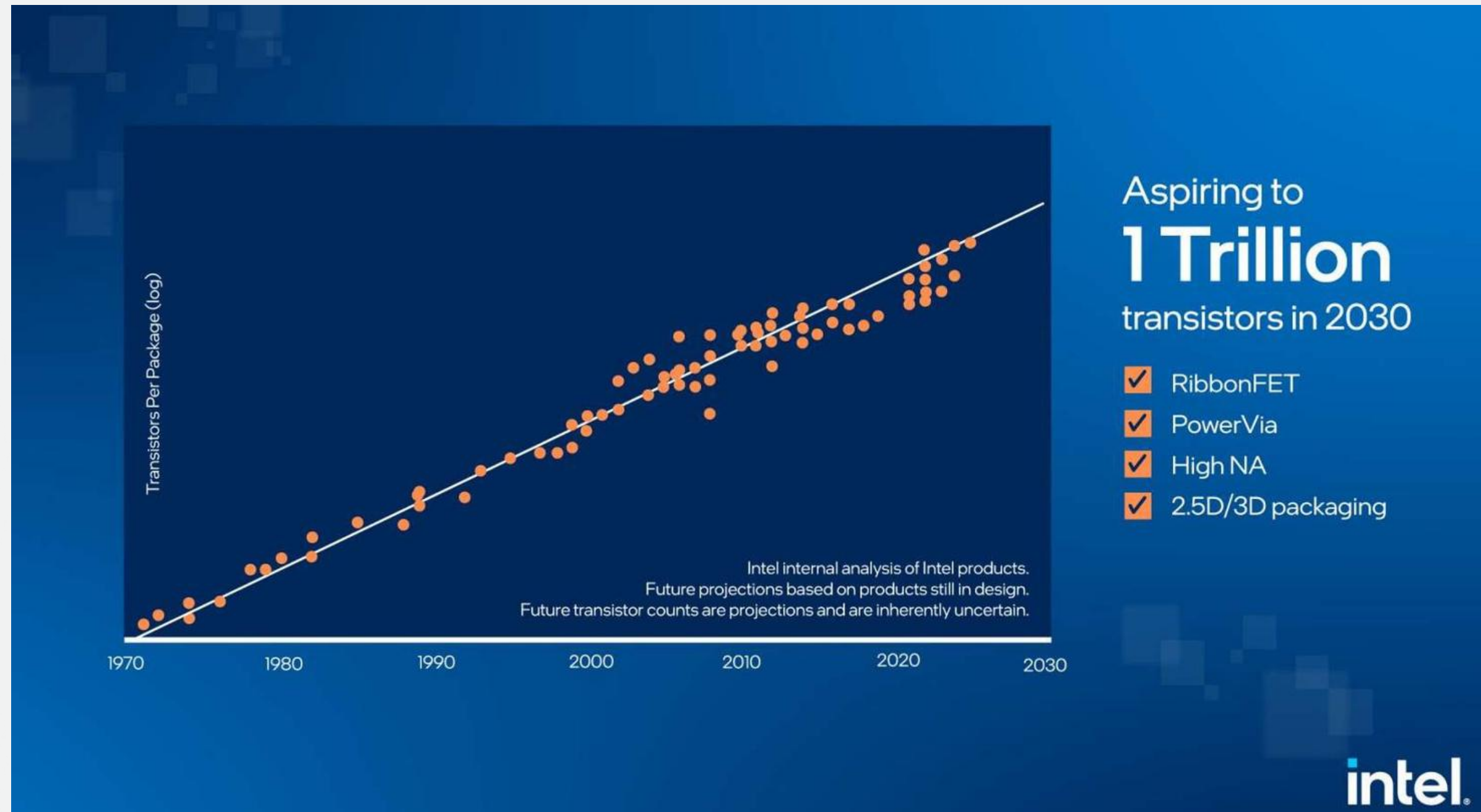
$$\text{Slope} = \frac{d(\log I_{ds})}{dV_{gs}} = \frac{1}{S} \quad \text{"S" = Subthreshold Swing}$$

- VELOCITY SATURATION OF CARRIERS

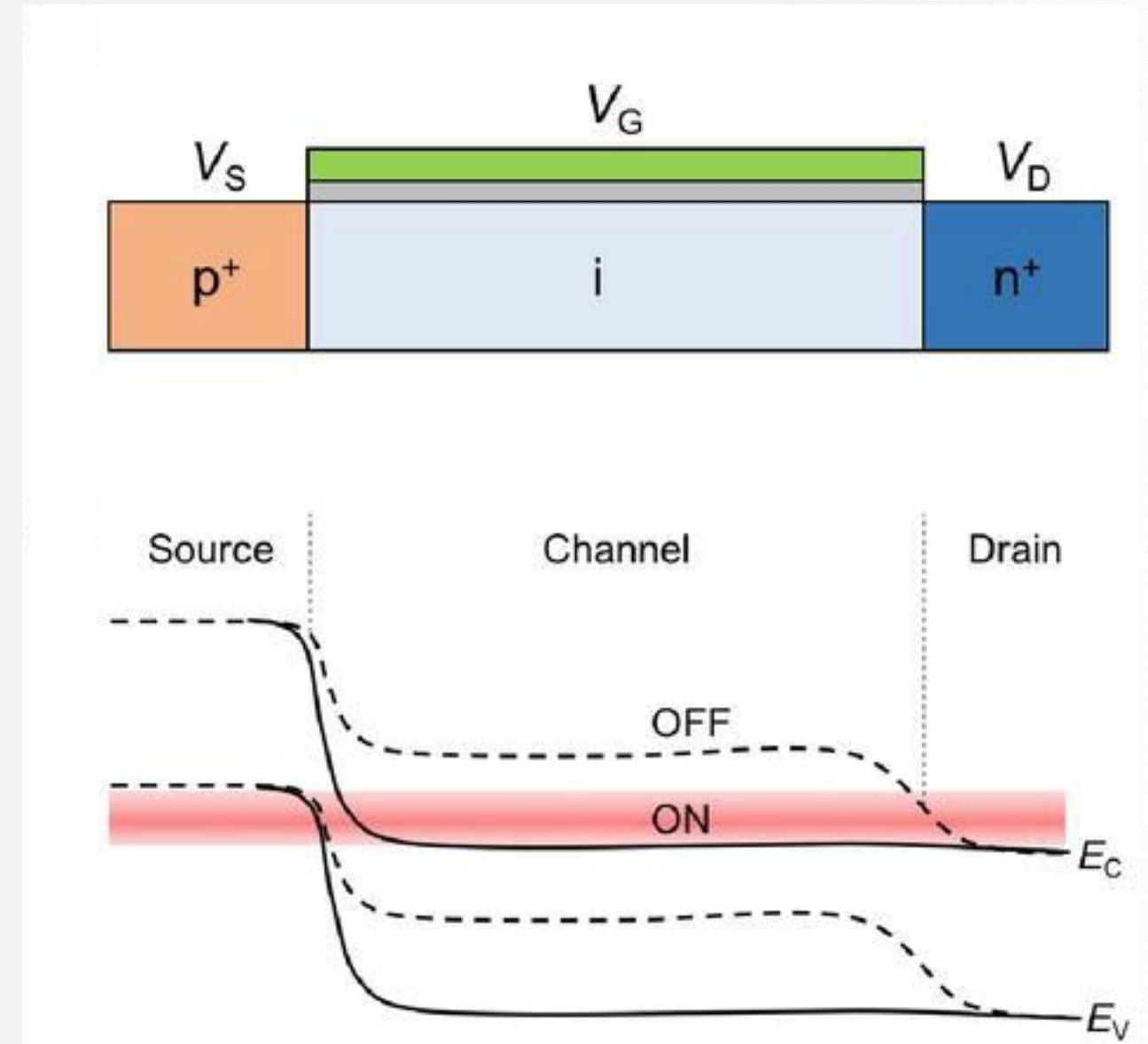
WHY CAN'T WE REDUCE THE SIZE OF THE MOSFETS



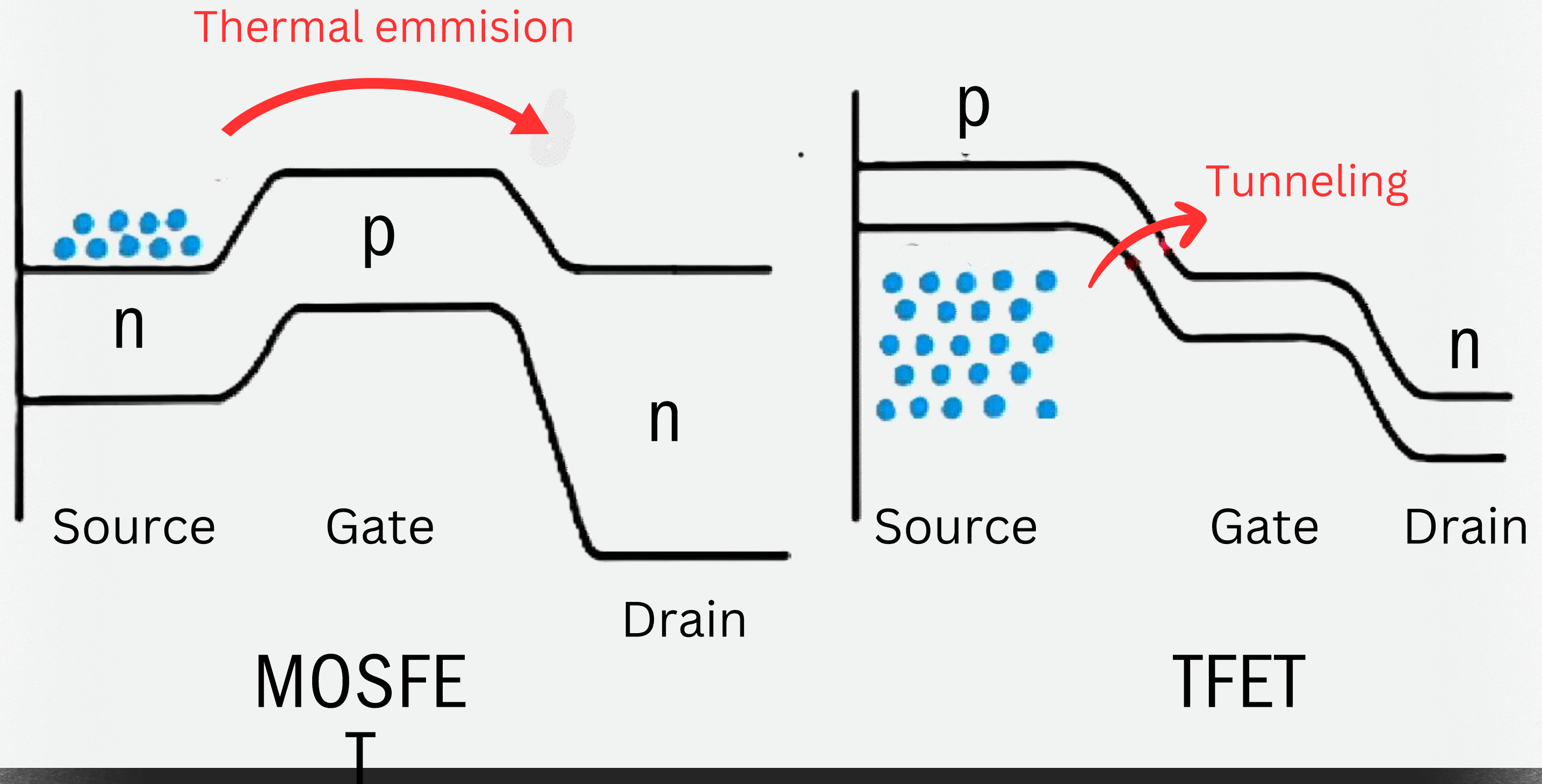
MOORE'S LAW



TUNNELING FIELD EFFECT TRANSISTOR

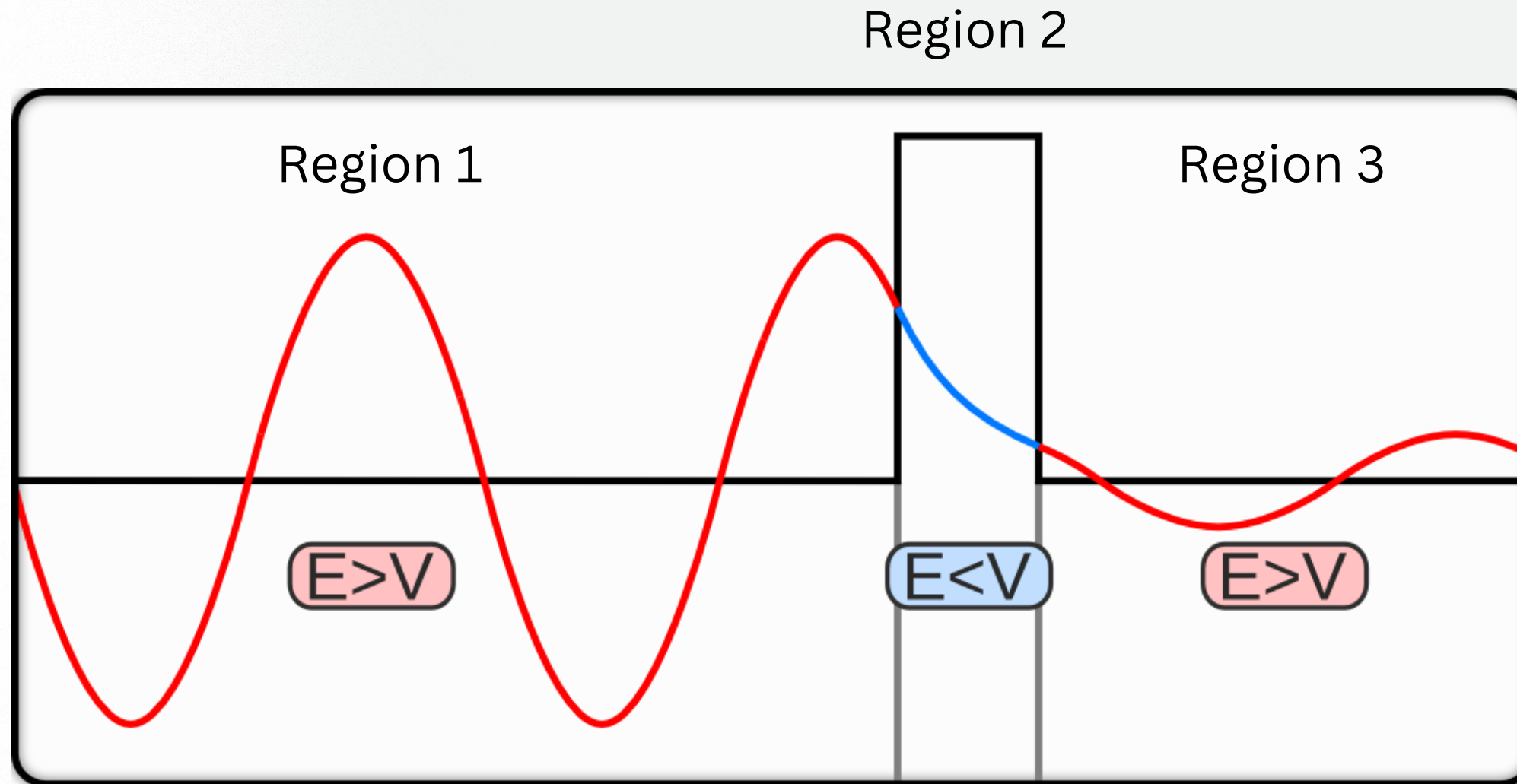


TFET MECHANISM



EC

1D TUNNELING



1. **Region 1** ($-\infty < x < 0$): $E > V(x)$
2. **Region 2** ($0 \leq x \leq L$): $E < V(x)$
3. **Region 3** ($L < x < \infty$): $E > V(x)$

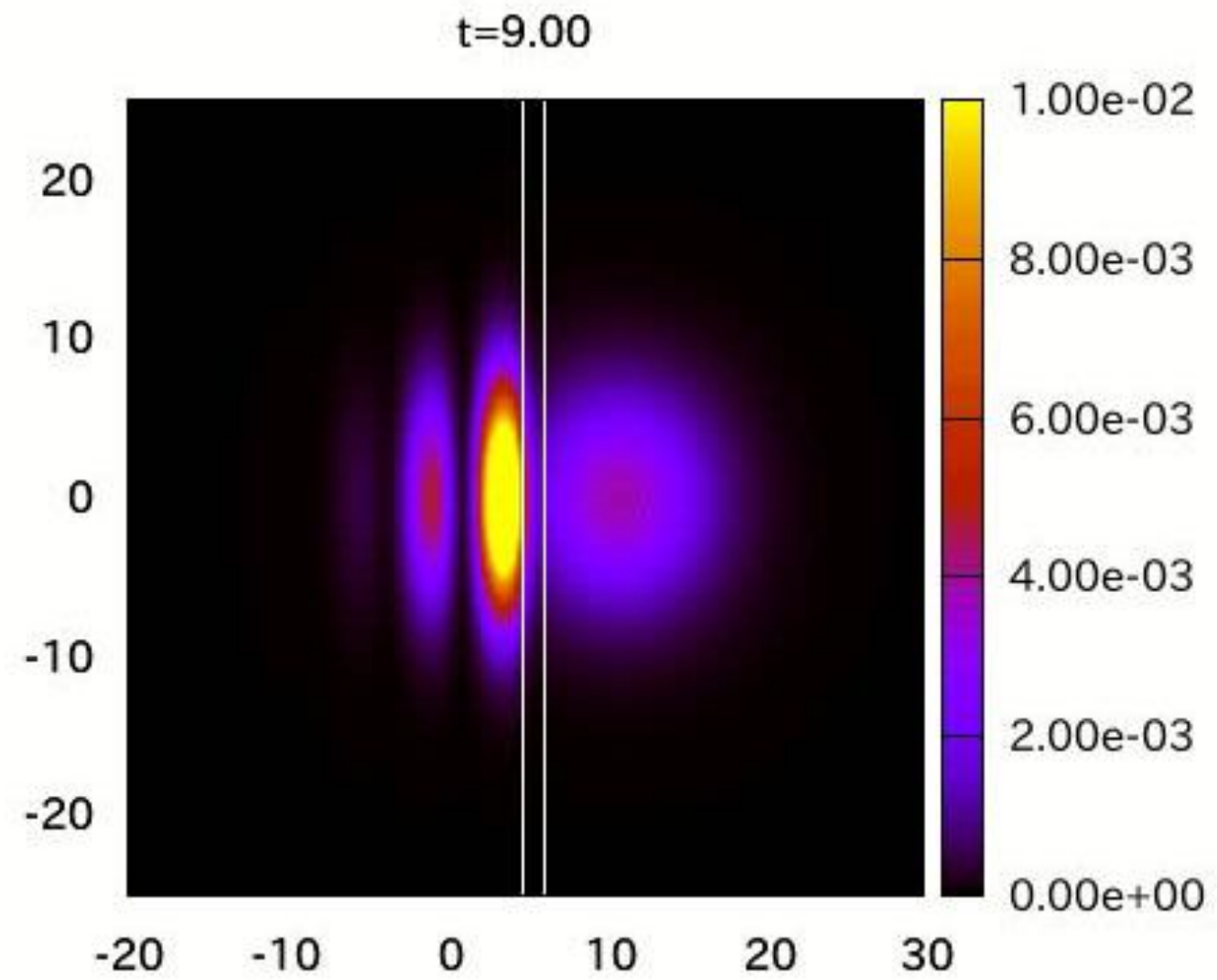
$$-\frac{\hbar^2}{2m} \frac{d^2 \psi(x)}{dx^2} + V(x) \psi(x) = E \psi(x)$$

1D TUNNELING

$$\psi(x) = \begin{cases} Ae^{ikx} + Be^{-ikx} & : \text{Region 1} \\ Ce^{\kappa x} + De^{-\kappa x} & : \text{Region 2} \\ Fe^{ikx} & : \text{Region 3.} \end{cases}$$

PROBABILITY TO FIND AND ELECTRON IN REGION 3 = $|\Psi_3|^2 = F^2$

1D TUNNELING

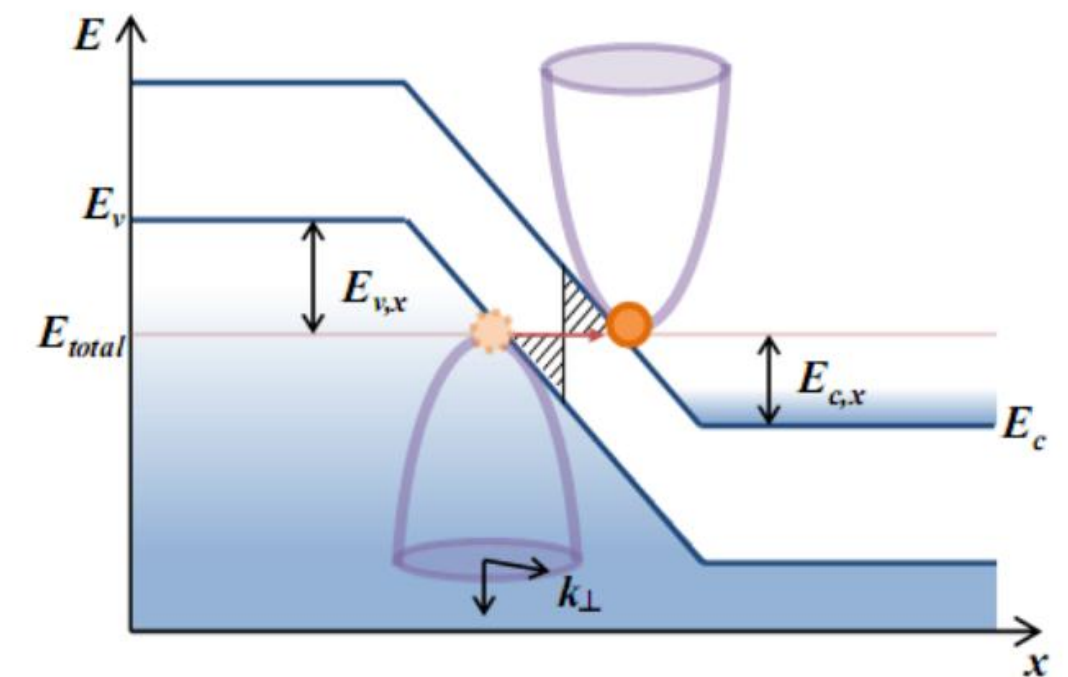


WKB APPROXIMATION

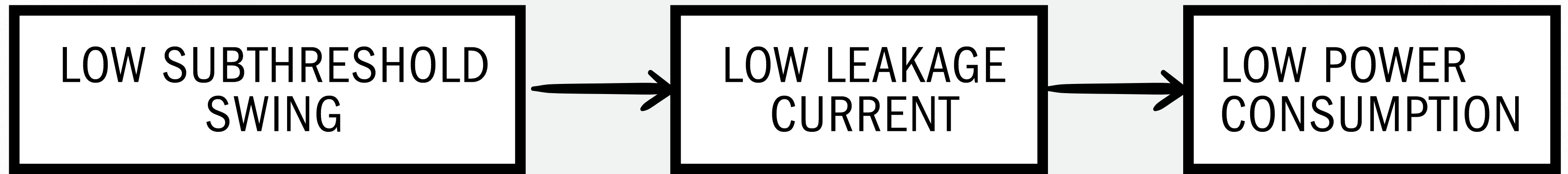
According to the WKB approximation,
transmission coefficient is

$$T \sim \exp \left[-2 \left(\int_a^{x_0} \kappa_{vx} dx + \int_{x_0}^b \kappa_{cx} dx \right) \right]$$

(b) E - k plot shown with Energy Band Diagram
for a Particle with $k_{y\perp}, k_{z\perp} = 0$



ADVANTAGES OF TFET



CHALLENGES OF TFET

- THE ON CURRENT IS NOT SUFFICIENT FOR THE ITRS VALUE
ITRS - (INTERNATIONAL TECHNOLOGY ROADMAP FOR SEMICONDUCTORS)
- FABRICATION COMPLEXITY

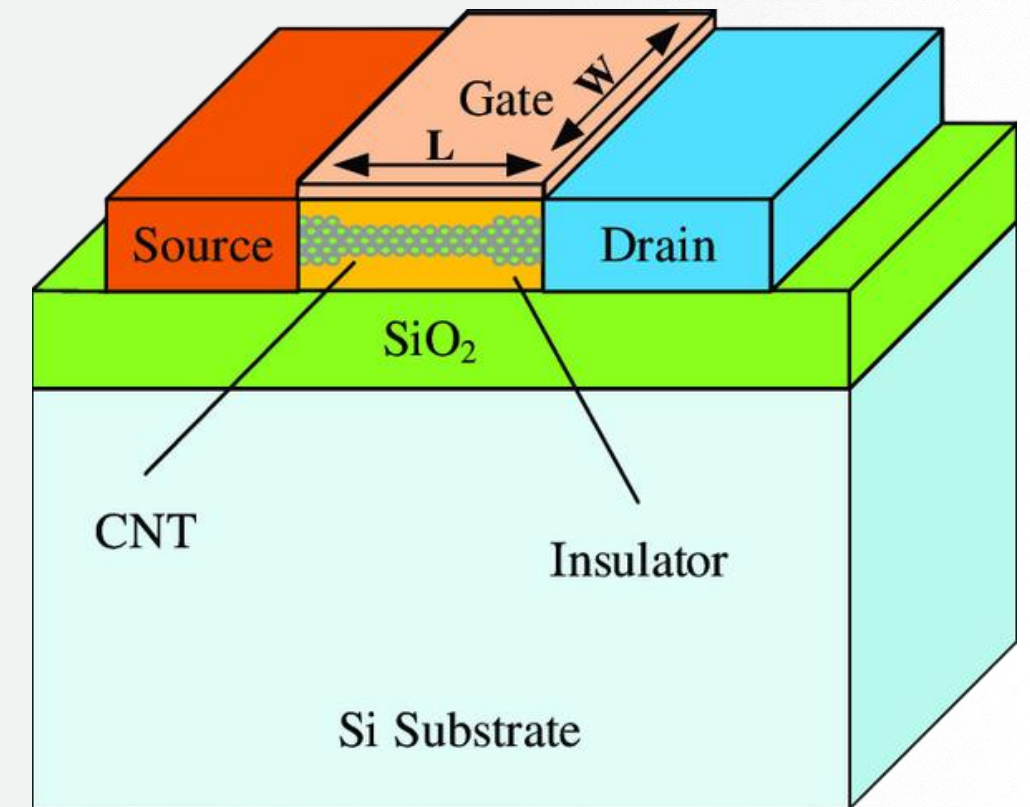
FUTURE DEVELOPMENTS

Carbon Nanotube TFET

- PROVIDE MULTIPLE SINGLE WALL CNTS AS TUNNELING PATH.
- MAIN ADVANCEMENT: -
 - CAN TRANSFER HIGH DRAIN TO SOURCE CURRENT THAN TFET.
 - CNTFET HAS LOW SUBTHRESHOLD SWING .

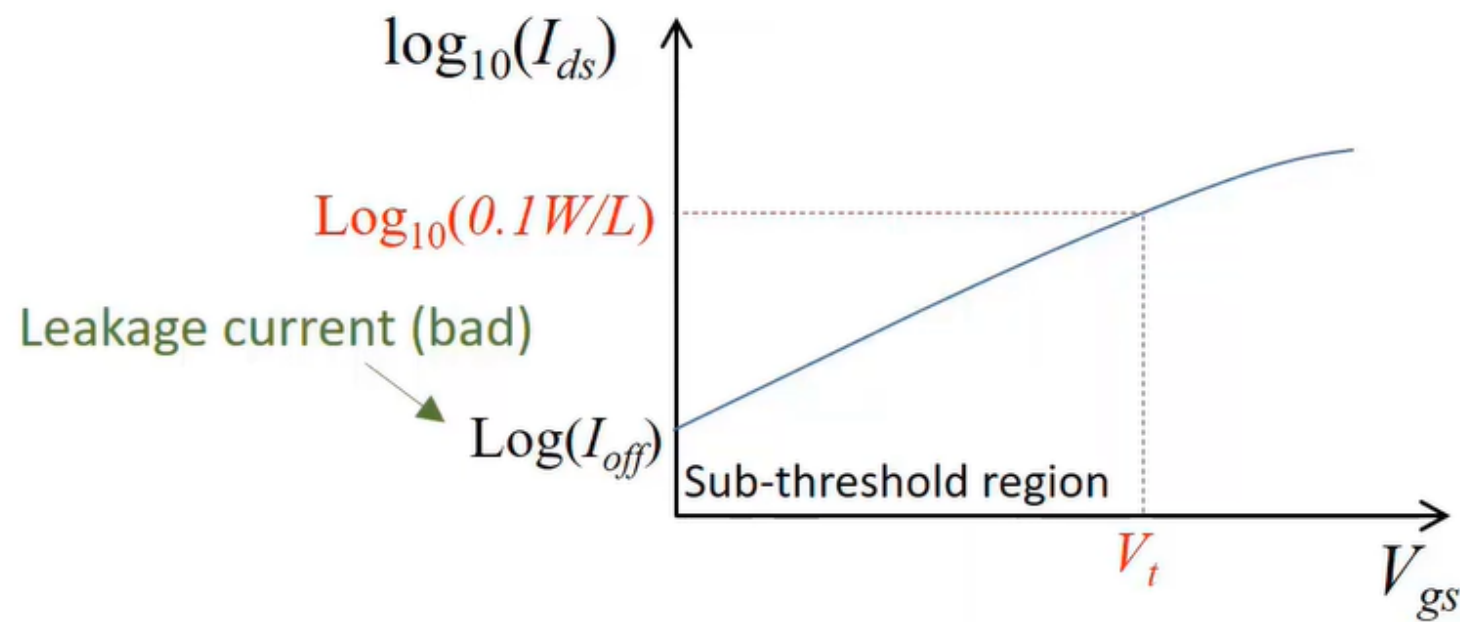
Heterojunction TFET

- TO SHORTEN THE TUNNELING DISTANCE TO INCREASE THE ON CURRENT AND ALSO TO REGULATE THE AMBIPOLAR LEAKAGE





**THANK
YOU**



$$I_{ds} = 0.1 \frac{W}{L} e^{-qV_t/\eta kT} e^{qV_{gs}/\eta kT}$$

$$I_{off} = 0.1 \frac{W}{L} e^{-qV_t/\eta kT} \quad [\mu\text{A}]$$

V_{gs}	$e^{qV_{gs}/\eta kT} = e^{V_{gs}/\eta 0.026}$
(100 mV) η	47
(160 mV) η	470
(220 mV) η	4,700
(280 mV) η	47,000

$\exp(qV_{gs}/\eta kT)$ [and consequently I_{ds}] changes by 10x for every **(60 mV) η** change in V_{gs} .

$$\downarrow \mu \propto \frac{1}{E} \uparrow$$

$$\nu = \mu' E$$

Sr. No.	Device	I_{on} (A/ μ m) ↑	I_{off} (A/ μ m) ↓	Subthreshold Swing (SS) (mV/dec) ↓		Transcon ductance (gm) (S) ↑	PDP (J/ μ m) ↓	Time Delay (s) ↓
				$V_{dd}=0.3$ V	$V_{dd}=1.2$ V			
1	MOSFET	128.714e-05	336348e-12	175.57	230	4907e-7	3.922e-16	2.53894e-13
2	Si TFET	4.68133e-05	0.255896e-12	50.41	214	7.97e-07	3.183e-17	5.666235e-13
3	Ge TFET	9.73722e-05	45.2214e-12	71.76	329	27.52e-07	9.449e-16	80.86858e-12
4	Si _{0.65} Ge _{0.35} TFET	6.99737e-05	1.48351e-12	57.56	227	25.74e-07	1.036e-15	123.3240e-13